

Homework NLP Omangali Armaral 23B071564

$$x_1 = 0.35$$

$$x_2 = 0.9$$

$$w_{11} = 0.0991$$

$$w_{12} = 0.3971$$

$$w_{21} = 0.7176$$

$$w_{22} = 0.5928$$

$$w_{13} = 0.4749$$

$$w_{23} = 0.7525$$

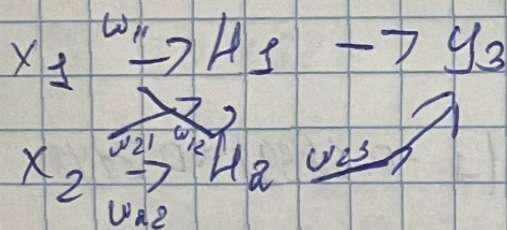
$$f(z) = \frac{1}{1 + e^{-z}}$$

$$(h_1)$$

$$z_1 = x_1 w_{11} + x_2 w_{21}$$

$$z_1 = (0.35)(0.0991) + (0.9)(0.7176)$$

$$z_1 = 0.034685 + 0.64585 \approx 0.75525$$



$$h_1 = f(z_1) = \frac{1}{1 + e^{-0.75525}} \approx 0.6800$$

$$(h_2) \quad z_2 = x_1 w_{12} + x_2 w_{22}$$

$$(X) \quad z_2 = (0.35)(0.3971) + (0.9)(0.5928)$$

$$z_2 = 0.138985 + 0.53352 \approx 0.67250$$

$$h_2 = f(z_2) = \frac{1}{1 + e^{-0.67250}} \approx 0.6620$$

$$z_3 = h_1 w_{13} + h_2 w_{23}$$

$$z_3 = (0.6800)(0.4749) + (0.6620)(0.7525)$$

$$z_3 \approx 0.3229 + 0.4988 = 0.8217$$

$$y_3 = f(z_3) = \frac{1}{1 + e^{-0.8217}} \approx 0.6889$$

Выходной нейрон

ошибка :

$$y_t = 0.5$$

$$\sum_{\text{error}} = y_t - y_3 = 0.5 - 0.6889 = -0.1889$$

Iteration 1.

$$(y) \quad f'(y) = y(1-y)$$

$$(y_t) \quad y_t = 0.5$$

$$\delta_3 = (y_t - y_3) f'(y_3)$$

$$f'(y_3) = y_3(1-y_3) = 0.6889(1-0.6889) \approx 0.2148$$

↓ произв. активации нейрона.

$$\sum_{\text{error}}: y_t - y_3 = 0.5 - 0.6889 = -0.1889$$

$$\delta_3 = (-0.1889)(0.2148) \approx -0.0406$$

$$(\text{h}_2) \quad \delta_2 = f'(h_2) (w_{23} \delta_3)$$

$$f(h_2) = 0.6800(1-0.6800) = 0.2176$$

$$w_{23} \delta_3 = 0.4749 (-0.0406) = -0.01928$$

$$\delta_1 = 0.2176 (-0.01928) \approx -0.00420$$

$$(h_2): \delta_2 = f'(h_2) (w_{23} \delta_3)$$

$$f'(h_2) = 0.6620(1 - 0.6620) = 0.2239$$

$$w_{23} \delta_3 = 0.7525(-0.0406) = -0.03055$$

$$\delta_2 = 0.2239(-0.03055) \approx -0.00684$$

$$\eta = 0.1$$

$$w_{13}: \Delta w_{13} = \eta \delta_3 h_1 = 0.1(-0.0406)(0.6800) = -0.0026$$

$$w_{13}^{new} = 0.4749 + \Delta w_{13} \approx 0.4723$$

$$\Delta w_{23} = 0.1(-0.0406)(0.6620) = -0.00269$$

$$w_{23}^{new} = 0.7526 - 0.00269 = 0.7498$$

↓

for h_3 .

$$\Delta w_{11} = 0.1(\delta_3 x_1) = 0.1(-0.00429)(0.35) = -0.000147$$

$$w_{11} = 0.09911 - 0.000147 = 0.09895$$

$$\Delta w_{21} = 0.1(-0.00429)(0.9) = -0.000378$$

$$w_{21}^{new} = 0.7546 - 0.000378 = 0.7177 \quad \rightarrow \text{for } h_2$$

$$w_{12} = 0.11(\delta_2 x_1) = 0.1(-0.00684)(0.35) = -0.000239$$

$$w_{12}^{new} = 0.3973 - 0.000239 = 0.39686$$

$$w_{22} = 0.1(\delta_2 x_2) = 0.1(-0.00684)(0.9) = -0.0006156$$